

Worked Solutions

Data Analytics

Set B: Original | CSE Semester 3

Companion to da-original.pdf. Question numbers match that paper. Equivalent correct methods are acceptable. These are independent worked solutions, not an official university marking scheme.

Questions 1–5: MCQ answer key, 1 mark each (5 marks total). Questions 6 onward: theory solutions (25 marks total).

1. Arithmetic mean

Correct option: (C). The sum is 48 and $48/6 = 8$.

2. Median

Correct option: (B). The middle values are 6 and 8, so the median is $(6 + 8)/2 = 7$.

3. Population variance

Correct option: (A). Population variance divides the squared-deviation sum by N : $64/6$.

4. Pearson skewness sign

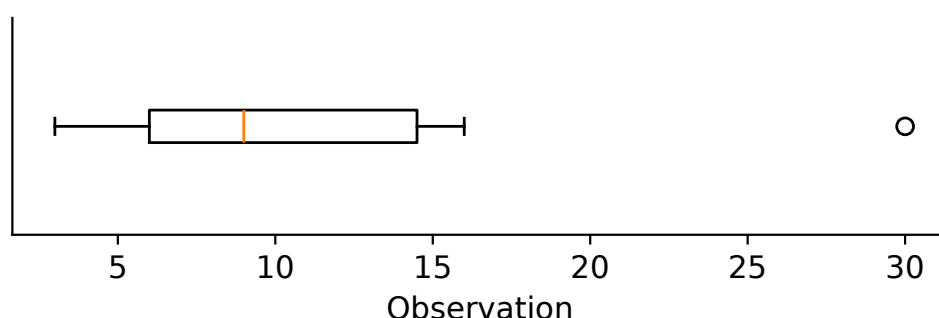
Correct option: (D). $3(\bar{x} - \text{median})/\sigma = 3(8 - 7)/\sigma > 0$.

5. Resistance to extremes

Correct option: (B). The median depends on the middle ranks and is generally more resistant to extremes.

6. Five-number summary and outlier

The median is 9. Excluding it gives lower half (3, 5, 7, 8) and upper half (11, 13, 16, 30). Thus $Q_1 = 6$, $Q_3 = 14.5$, and the five-number summary is (3, 6, 9, 14.5, 30). $IQR = 8.5$; the fences are -6.75 and 27.25 . Therefore 30 is an outlier. The modified box plot has whiskers at 3 and 16, with 30 plotted separately.



7. Missing and inconsistent data

Impute missing ages with the median of a relevant, sufficiently large customer group and retain a missingness flag. Estimate imputation values from training data only if building a predictive model. Flag rows where annual income differs materially from $12 \times$ monthly income after aligning currency, units, gross/net definitions and time period. Investigate such rows rather than automatically overwriting them.

8. Filling-machine mean test

$H_0 : \mu = 500$ ml, $H_1 : \mu \neq 500$ ml. The standard error is $12/\sqrt{36} = 2$ ml.

$$z = \frac{496 - 500}{2} = \boxed{-2.00}.$$

Since $|-2| > 1.96$, reject H_0 at 5%. There is evidence that the machine's mean fill differs from 500 ml; the sample suggests underfilling. The two-sided p -value is about 0.0455.

9. Rank correlation

The six rank differences are $(-1, 1, -1, 1, -1, 1)$, giving $\sum d_i^2 = 6$.

$$r_s = 1 - \frac{6(6)}{6(6^2 - 1)} = \boxed{\frac{29}{35} = 0.8286}.$$

There is strong positive rank agreement, although each adjacent pair is reversed.

10. Equal-frequency smoothing

Bin	Observations	Mean
1	10,12,15,18	$55/4 = 13.75$
2	20,22,25,28	$95/4 = 23.75$
3	30,32,35,40	$137/4 = 34.25$

The smoothed sequence is four copies of 13.75, followed by four copies of 23.75 and four copies of 34.25.

11. Training and passing

H_0 : training and passing are independent. Row totals are 60 and 40; column totals are 70 and 30. Expected counts are

$$E = \begin{pmatrix} 42 & 18 \\ 28 & 12 \end{pmatrix}.$$

$$\chi^2 = \frac{(48 - 42)^2}{42} + \frac{(12 - 18)^2}{18} + \frac{(22 - 28)^2}{28} + \frac{(18 - 12)^2}{12} = \boxed{7.1429}.$$

With one degree of freedom, $7.1429 > 3.841$, so reject independence. Training and passing are associated. The pass rates are 80% and 55%; causation requires an appropriate study design.

12. One-way ANOVA

Source	SS	df	MS	F
Between	54	2	27	27
Within	6	6	1	
Total	60	8		

Test $H_0 : \mu_A = \mu_B = \mu_C$. Since $F = 27 > 5.143$, reject H_0 at 5%. At least one population mean differs. Pairwise differences require further testing.

13. Type I and Type II errors

A Type I error is rejecting $\mu = 500$ when the machine's true mean is 500 ml. A Type II error is failing to reject $\mu = 500$ when the true mean differs from 500 ml. The 5% significance level controls the Type I error probability under the test assumptions; it does not specify the Type II probability.